

Dual-Channel Attention: Critical Phase Transitions and Trail Fields in Neural Sequence Processing

Research Notes

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Abstract

Systems poised at a critical phase transition—between quiescent and runaway-active regimes—exhibit maximal dynamic range, information transmission, and adaptability. Neuronal avalanches in cortex and pheromone trail formation in ant colonies both belong to the directed percolation universality class, governed by the same three minimal ingredients: *amplification* (positive feedback), *injection* (activity-dependent deposition), and *erosion* (uniform decay)—subject to the constraint, demonstrated in a companion lattice ant study, that erosion must be *uncoupled* from the field it erodes. We ask whether importing these ingredients and this constraint into transformer attention produces measurable computational benefits. We propose *dual-channel attention*, which decomposes the attention mechanism into a *structured channel* (content-based Q/K matching) and a *social channel*: a per-head vector recurrence whose output modulates the key geometry of attention, reshaping *what* the model matches against rather than merely biasing *how much* attention each position receives. The social channel implements all three critical ingredients within the attention computation itself, providing intrinsic sequential memory that is architecturally distinct from content matching. We ground this design in the neurobiology of synaptic tagging, volume transmission, and activity-dependent myelination—mechanisms that are functionally stigmergic—and frame the architecture as an exploration of criticality in learned sequence processing.

1 Introduction

1.1 Criticality in Learning Systems

A growing body of evidence suggests that biological neural systems operate near a *critical phase transition*—the boundary between a quiescent regime (activity dies out) and a runaway-active regime (activity explodes). At this critical point, systems exhibit maximal dynamic range, maximal information transmission, and maximal sensitivity to input [1, 2, 3].

Beggs and Plenz [1] showed that spontaneous activity in cortical slices produces *neuronal avalanches* whose size distribution follows a power law with exponent $-3/2$ —the signature of the *mean-field directed percolation* (DP) universality class. The branching parameter σ converges to ~ 1 : below 1, activity dies; above 1, it explodes; at exactly 1, it persists at all scales.

This is not unique to neurons. Munoz [4], in a comprehensive review in *Reviews of Modern Physics*, shows that the same critical scaling laws appear across living systems—from gene regulatory networks to bird flocks to insect colony foraging. The Janssen–Grassberger conjecture [5, 6] provides the theoretical bridge: *any* system with (1) a continuous phase transition into an absorbing state, (2) a one-component order parameter, (3) short-range interactions, and (4) no additional symmetries belongs to the DP universality class.

1.2 The Three Ingredients

The simplest lattice model in the DP universality class is the *contact process* [7]: sites on a grid are active or inactive; active sites activate neighbors at rate λ and deactivate at rate 1. The all-inactive state is absorbing. At critical λ_c , the system undergoes a continuous phase transition with DP exponents.

What are the minimal ingredients for this transition? Following Deneubourg et al. [9] and the ant trail literature, we identify three:

1. **Amplification** (positive feedback): activity at a site increases the probability of future activity at that site or its neighbors.
2. **Injection** (content-dependent deposition): active sites deposit a trace into a shared medium—a signal that persists beyond the lifetime of the activity that created it.
3. **Erosion** (uniform decay): traces fade at a constant rate, preventing runaway accumulation and enabling the system to forget.

These three ingredients are necessary and sufficient for the DP phase transition. Remove any one and the transition vanishes: without amplification, traces have no effect; without injection, there is nothing to amplify; without erosion, the system saturates.

A crucial additional constraint, revealed by systematic study of our lattice ant model, is that **erosion must be uncoupled from the field it erodes**. Of nine model variants tested, only those with uniform, field-independent decay produced edge-of-chaos dynamics. Every variant that introduced decay or noise *correlated* with the pheromone concentration—follow-probability modulation, habituation, sensory saturation, coupled deposition—destroyed the critical ridge entirely. The mechanism is intuitive: field-correlated perturbations feed back into the amplification loop, creating a self-cancelling cycle that prevents the system from reaching the delicate balance between growth and decay that defines criticality. In stigmergic systems, the hard problem is not *producing* edge-of-chaos dynamics but *not destroying* them. We therefore treat **field-independent erosion** as a fourth design requirement, equal in importance to the three ingredients themselves.

1.3 Ant Trails as the Simplest Illustration

Ant trail formation is the most intuitive system exhibiting these ingredients. Ants deposit pheromone (injection), subsequent ants preferentially follow stronger trails (amplification), and pheromone evaporates (erosion). Deneubourg’s bifurcation model [9] shows that above a critical interaction frequency, symmetry breaks and coherent trails emerge—a continuous phase transition with the same mathematical structure as neuronal avalanches.

We emphasize: the ant model is not the *motivation* for our architecture. It is the simplest system that makes the critical ingredients visible. The motivation comes from the shared universality class connecting ant dynamics and neural dynamics.

1.4 Neural Biological Parallels

Biological neurons possess mechanisms that are functionally stigmergic—activity leaves traces in a shared medium that modulate future activity:

- **Synaptic tagging** [10]: A weak stimulus creates a short-lived, protein-synthesis-independent “tag” at a synapse. A strong stimulus at *any* tagged synapse triggers plasticity-related protein synthesis captured by *all* tagged synapses. Tags decay without reinforcement (erosion), activity creates them (injection), and they bias future plasticity (amplification).

- **Nitric oxide volume transmission** [11, 12]: NO diffuses isotropically through tissue, affecting neurons with no active synaptic connection to the source. This is four-dimensional environmental signaling—a diffuse field, not a directed synapse.
- **Perineuronal nets**: Extracellular matrix structures surrounding neurons are modified by experience, physically constraining future plasticity—a literal environmental trace.
- **Activity-dependent myelination**: Heavily used axons receive more myelin, increasing conduction speed. Use modifies the physical substrate, which modifies future signal propagation—trail reinforcement.

None of these mechanisms are typically called “stigmergic” in the neuroscience literature, but they satisfy the definition precisely: indirect coordination through modification of a shared environment. The term *stigmergy* (Grassé, 1959) was coined for termite construction, but the mathematical structure—*injection, amplification, erosion* leading to a DP-class phase transition—appears to be a universal motif in biological information processing.

1.5 From Criticality to Architecture

Our hypothesis: importing the three critical ingredients—plus the uncoupling constraint—into transformer attention should produce measurable computational benefits, particularly for tasks requiring sequential state maintenance—where standard attention’s memoryless query/key matching is provably limited.

In a companion paper [13], we tested this with the simplest possible implementation: a *scalar* pheromone field biasing attention logits. The results established that the feedback loop matters (deposits must come from attention *output*, not input, to close the amplification loop), but the scalar field proved too weak. The effect was directional: +0.07 accuracy on parity length generalization, modest improvement on flipflop, no effect on cumulative sum. A scalar field decoupled from the attention geometry cannot express state-dependent routing—it can only say “attend more here,” not “attend *differently* here.” Softmax normalization absorbs most of the scalar signal.

This paper proposes *dual-channel attention*: a vector social field that modulates the *geometry* of key space, not just the magnitude of attention scores. The field operates in the same representational space as Q/K matching, making it a co-equal routing mechanism rather than a decorative bias. Crucially, the social field’s decay is *uniform*—a fixed retention factor α applied identically regardless of the field’s current magnitude—in accordance with the uncoupling constraint. Content-dependent gating controls what is *deposited*; erosion remains content-blind.

2 Why Scalar Biases Are Insufficient

Consider two positions j_1, j_2 with raw attention scores s_1, s_2 and scalar field values ϕ_1, ϕ_2 . The relative attention ratio is:

$$\frac{a_{j_1}}{a_{j_2}} = \frac{\exp(s_1 + \phi_1)}{\exp(s_2 + \phi_2)} = \frac{\exp(s_1)}{\exp(s_2)} \cdot \exp(\phi_1 - \phi_2) \quad (1)$$

Only the *difference* $\phi_1 - \phi_2$ matters. If the field encodes a running state (e.g., parity), all positions receive similar accumulated values, making the difference small and the bias negligible.

A scalar bias per position can only say “attend more here” or “attend less here.” It cannot say:

- “The sequence so far contains an odd count—match against keys that encode parity-sensitivity.”

- “The last state-changing event was a SET 5 command—adjust the key space to surface positions carrying that register value.”

These require modulating the *geometry* of attention, not its *magnitude*. A vector field that modulates keys changes which *dimensions* of the key space matter, creating orthogonal routing channels that softmax normalization cannot absorb.

3 Why Start with Attention

The three critical ingredients could in principle be added to any differentiable computation: MLPs, convolutions, SSMs. We target attention specifically for three reasons.

First, attention is *memoryless between positions*. Each position computes its output from scratch via Q/K matching, with no carried state. This is precisely the property that makes standard attention fail at sequential state-tracking tasks (parity, counting, register maintenance)—tasks where biological neural systems, which *do* have environmental traces (synaptic tags, NO fields), succeed. Attention is where the gap between biological and artificial computation is largest.

Second, attention already has the *amplification* ingredient in nascent form. The softmax nonlinearity creates winner-take-all dynamics: small score differences produce large attention differences. What is missing is *injection* (activity leaving a persistent trace) and *erosion* (traces decaying). Adding these two ingredients to a system that already amplifies should be the minimal intervention needed to reach the critical regime.

Third, attention has a natural *geometric structure*—the Q/K dot-product space—that a social field can modulate. Unlike scalar activations in MLPs, attention keys are vectors in a learned representation space. A vector social field can reshape this space, creating content-and-history-dependent routing that operates in the same representational language as the structured lookup. This is qualitatively different from adding a recurrence alongside attention (as in Mamba or RWKV): the field is *inside* the attention computation, not parallel to it.

4 Three Architectures

Standard attention is purely *individual*: each position performs its own content-based lookup, with no memory of what prior positions found useful. In the ant analogy, this is an ant following only its own sensory gradient—functional, but ignorant of the colony’s experience.

We augment attention with a *social field*: a shared, decaying, vector-valued signal that accumulates deposits from attention outputs and feeds back into future attention computations. The social field is the colony’s pheromone—an environmental medium that carries the collective processing history of the sequence. Each position reads the social field (what has worked before?) and writes to it (what did I find?), creating the feedback loop required for criticality.

We propose three architectures that implement this social field within attention, ordered by expressivity. All three share the same field recurrence but differ in how the field couples to the attention geometry. Together they form a progression from the minimal intervention to a fully content-dependent modulation of the key manifold.

4.1 Shared Component: The Social Field

Definition 1 (Social field). *For head h , the social field at position i is a vector $\phi_i^{(h)} \in \mathbb{R}^{d_f}$ defined by the recurrence:*

$$\phi_i^{(h)} = \alpha \cdot \phi_{i-1}^{(h)} + d_i^{(h)}, \quad \phi_{-1}^{(h)} = \mathbf{0} \quad (2)$$

where:

- $\alpha = 1 - \varepsilon \in (0, 1)$ is the retention rate (erosion—the field forgets without reinforcement)
- $\mathbf{d}_i^{(h)} \in \mathbb{R}^{d_f}$ is the deposit at position i , computed from the attention output (injection—processing leaves a trace):

$$\mathbf{d}_i^{(h)} = W_{dep}^{(h)} \mathbf{o}_i \quad (3)$$

with $W_{dep}^{(h)} \in \mathbb{R}^{d_f \times d}$ and $\mathbf{o}_i$ the attention output at position i

- d_f is the field dimension, controlling the information capacity of the social channel

The field implements two of the three critical ingredients directly. Amplification—the third ingredient—arises from the coupling: the field modulates attention, which changes the output, which changes the deposit, which changes the field.

The term *social* is deliberate. In attention, queries and keys are *individual*—each position’s private representation. The social field is *collective*—it belongs to no single position but accumulates the processing history of the entire sequence so far. An individual position reads the social field to discover what prior processing found useful and writes to it to inform future positions. This is precisely the stigmergic coordination loop: indirect communication through a shared, decaying environmental medium.

The field dimension d_f controls state capacity. In the scalar case ($d_f = 1$), this reduces to the stigmergic attention bias of our companion paper. We set $d_f = d_h$ (the head dimension) so the social field and keys live in the same representational space.

Remark 1 (State capacity). *Each head carries d_f continuous values of social state. With H heads, the total social state is $H \times d_f$ —compared to 0 for standard attention and n for SSMs with state dimension n . For $H = 4, d_f = 12$, this is 48 values of persistent state, vastly more than the $H = 4$ scalars of the stigmergic architecture.*

Remark 2 (Why modulate keys, not queries?). *Keys are the offerings of past positions; queries are the demands of the current position. The social field modulates offerings, not demands: it reshapes what information is available based on collective processing history. This preserves the individual nature of queries—each position’s sensory state remains purely content-driven—while making the key landscape socially informed. In the ant analogy: pheromone marks where ants have walked (keys), not where they want to go (queries).*

4.2 Architecture I: Additive Key Shift

The simplest proposal. The social field is projected to key dimension and added to keys before Q/K matching:

$$\mathbf{k}'_j^{(h)} = \mathbf{k}_j^{(h)} + W_{mod}^{(h)} \phi_j^{(h)} \quad (4)$$

where $W_{mod}^{(h)} \in \mathbb{R}^{d_h \times d_f}$.

Interpretation. The attention score decomposes cleanly into two channels:

$$\begin{aligned} \text{score}(i, j) &= \frac{1}{\sqrt{d_h}} \mathbf{q}_i^{(h)\top} \mathbf{k}'_j^{(h)} \\ &= \underbrace{\frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)}}{\sqrt{d_h}}}_{\text{structured (content match)}} + \underbrace{\frac{\mathbf{q}_i^{(h)\top} W_{mod} \phi_j^{(h)}}{\sqrt{d_h}}}_{\text{social (history match)}} \end{aligned} \quad (5)$$

Unlike the scalar bias of stigmergic attention, the social term is a *full dot product* between the query and a field-derived vector—it cannot be absorbed by softmax normalization because it varies across the d_h dimensions.

Parameters. W_{dep} ($d_f \times d$ per head) + W_{mod} ($d_h \times d_f$ per head). With $d_f = d_h$: $\sim 25\%$ overhead on the attention layer.

Why start here. This is the minimal intervention that gives the field access to the geometric structure of attention. If additive modulation shows no effect, the problem is not expressivity but something more fundamental (e.g., the sequential bottleneck prevents learning, or the tasks don’t benefit from social state).

4.3 Architecture II: Gating Modulation

The social field produces per-dimension gates that amplify or suppress key dimensions:

$$\mathbf{k}'^{(h)}_j = \mathbf{k}^{(h)}_j \odot \sigma(W_{\text{gate}}^{(h)} \phi_j^{(h)} + \mathbf{b}_g) \quad (6)$$

where σ is the sigmoid function and $\odot$ is elementwise multiplication.

Interpretation. Rather than shifting keys, gating *selects* which dimensions of the key space are active. The field controls which semantic features are matchable at each position. A gate near 0 on dimension k effectively removes that feature from the key; a gate near 1 preserves it. This is a multiplicative interaction—qualitatively stronger than additive because it can *silence* dimensions, not just nudge them.

Parameters. Same as additive plus a bias vector $\mathbf{b}_g \in \mathbb{R}^{d_h}$ per head. Negligible overhead beyond Architecture I.

When to prefer over I. Gating is appropriate when the task requires *selective attention*—attending to different aspects of keys depending on accumulated state. For example: in bracket matching, the field might gate out “nesting depth” dimensions when the stack is empty and gate them in when brackets are open.

4.4 Architecture III: Rotational Modulation

The field produces a content-and-state-dependent rotation of the key vector:

$$\mathbf{k}'^{(h)}_j = \mathbf{k}^{(h)}_j + W_{\text{rot}}^{(h)}(\phi_j^{(h)} \otimes \mathbf{k}^{(h)}_j) \quad (7)$$

where $\otimes$ denotes a bilinear interaction (outer product projected back to $\mathbb{R}^{d_h}$) and $W_{\text{rot}}^{(h)} \in \mathbb{R}^{d_h \times d_f \cdot d_h}$.

Interpretation. This is the most expressive variant. The modulation depends on *both* the field state and the key content—different keys at the same position are rotated differently. This creates a content-and-history-dependent routing that can implement arbitrary learned transformations of the key manifold.

Parameters. W_{rot} is $d_h \times (d_f \cdot d_h)$ per head, which is $O(d_h^2 \cdot d_f)$. This is expensive: for $d_h = d_f = 12$, it is $12 \times 144 = 1728$ parameters per head versus 144 for additive. A low-rank factorization $W_{\text{rot}} = UV^\top$ with rank $r \ll d_f$ can reduce this.

When to prefer over I–II. Rotational modulation is appropriate when the task requires the field to implement different transformations depending on key content—e.g., when the same accumulated state should affect “parity-encoding” keys differently from “register-encoding” keys. We expect this to matter primarily at scale, where different heads specialize.

4.5 Shared Forward Pass

All three architectures share the same forward pass structure. At each position i :

1. **Project** input to queries, keys, values: $\mathbf{q}_i, \mathbf{k}_i, \mathbf{v}_i = \text{QKV}(\mathbf{x}_i)$
2. **Read social field**: retrieve accumulated $\phi_j^{(h)}$ for all $j < i$
3. **Modulate keys**: $\mathbf{k}_j'^{(h)} = f(\mathbf{k}_j^{(h)}, \phi_j^{(h)})$ using one of the three strategies
4. **Attend**: compute attention weights using modulated keys

$$a_{ij}^{(h)} = \text{softmax}_j \left(\frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j'^{(h)}}{\sqrt{d_h}} \right) \quad j \leq i \quad (8)$$

5. **Aggregate**: $\mathbf{o}_i^{(h)} = \sum_{j \leq i} a_{ij}^{(h)} \mathbf{v}_j^{(h)}$
6. **Deposit**: $\mathbf{d}_i^{(h)} = W_{\text{dep}}^{(h)} \mathbf{o}_i^{(h)}$ (injection)
7. **Update field**: $\phi_i^{(h)} = \alpha \cdot \phi_{i-1}^{(h)} + \mathbf{d}_i^{(h)}$ (erosion)

The feedback loop closes the amplification ingredient:

$$\phi \xrightarrow{\text{modulate } K} \text{attention} \xrightarrow{\text{aggregate}} \text{output} \xrightarrow{W_{\text{dep}}} \text{deposit} \xrightarrow{\text{accumulate}} \phi \quad (9)$$

	I: Additive	II: Gating	III: Rotational
Modulation type	shift	selection	rotation
Field-key interaction	linear	multiplicative	bilinear
Extra params / head	$d_h d_f + d_f d$	same + d_h	$d_h^2 d_f + d_f d$
Content-dependent	no	no	yes
Can silence dimensions	no	yes	yes
Expressivity	low	medium	high

Table 1: Summary of the three proposed architectures. All share the same social field recurrence and forward pass; they differ only in how the field modulates keys. We recommend starting experiments with Architecture I (additive) as the simplest test of whether social fields provide computational benefit.

4.6 The Individual/Social Decomposition

Standard attention performs a single type of lookup—purely individual, purely content-based:

$$\text{score}(i, j) = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_h} \quad (\text{“what content at } j \text{ matches my query?”}) \quad (10)$$

All three architectures decompose this into an individual channel and a social channel. For Architecture I (additive), the decomposition is exact (cf. eq. (5)):

$$\text{individual}(i, j) = \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_h} \quad (\text{“what content matches?”}) \quad (11)$$

$$\text{social}(i, j) = \mathbf{q}_i^\top (W_{\text{mod}} \phi_j) / \sqrt{d_h} \quad (\text{“what has the sequence learned?”}) \quad (12)$$

The individual channel is memoryless: it depends only on the current query and the static key. The social channel carries history: it depends on the accumulated social field, which encodes what prior attention computations found productive. For Architectures II and III, the

decomposition is implicit but the principle is the same: one channel routes by content, the other by collective experience.

This is the core architectural claim: attention conflates individual and social information processing into a single dot product. Separating them—giving each its own mechanism, its own parameters, its own representational capacity—should allow the model to learn both more effectively.

5 Relationship to Existing Architectures

Architecture	Modulation	State	Feedback	Mechanism
Standard Transformer	none	—	—	Q/K matching
ALiBi [15]	logit bias	—	—	fixed distance penalty
T5 relative bias	logit bias	—	—	learned per-distance
RoPE [16]	query/key rotation	—	—	fixed rotation
Stigmergic (scalar)	logit bias	H scalars	nonlinear	output $\rightarrow$ scalar field
RetNet [17]	key decay	d	linear	γ -mask on keys
Mamba [18]	separate path	n	linear	selective SSM
RWKV [19]	separate path	d	linear	linear attention
Feedback Transformer	extra KV	d	nonlinear	top-layer output
Dual-channel	key modulation	$H \times d_f$	nonlinear	output $\rightarrow$ vector field

Table 2: Architectural comparison. Dual-channel attention is unique in using a *vector* recurrence to modulate *keys*, creating a content-dependent routing channel that operates in the same representational space as Q/K matching.

ALiBi and RoPE modify attention geometry but *statically*—the modification depends on position, not on content or processing history. Dual-channel attention is “ALiBi where the bias is a learned vector field driven by what attention has found useful.”

RetNet applies a multiplicative γ^{i-j} decay to keys, which is structurally similar to our decaying field. But RetNet’s decay is *fixed and content-independent*—it provides exponential forgetting but no feedback. The social field’s deposits are content-dependent and output-derived, creating adaptive decay.

Mamba and RWKV maintain recurrent state but in a *separate pathway* from attention. The SSM and attention do not interact—they are stacked, not coupled. In dual-channel attention, the recurrent state (social field) directly modulates the attention computation, and the attention output directly feeds the recurrence. The two mechanisms are co-dependent, not independent.

Feedback Transformer [20] is the closest prior work. It feeds the top layer’s output as an additional key-value pair for the next position. This provides output feedback but through a *single vector* added to the key-value set, not through modulation of *all* keys. The social field modulates every key at every position through a decaying recurrence, providing graded, history-dependent routing rather than a single feedback signal.

What is novel. The specific combination—a vector recurrence that modulates keys via additive/gating/rotational transforms, with deposits derived from attention output—does not appear in prior work to our knowledge. More fundamentally, the architecture is not motivated

by analogy but by universality: the three ingredients (amplification, injection, erosion) are the minimal recipe for DP-class phase transitions, and we import them into attention to test whether criticality confers computational advantages in learned sequence processing, as it does in biological neural systems [1, 2].

6 Analysis

6.1 Parameter Cost

For additive modulation with $d_f = d_h$:

- Deposit projection: $W_{\text{dep}} \in \mathbb{R}^{d_f \times d}$, adding $d_f \times d$ parameters per head
- Modulation projection: $W_{\text{mod}} \in \mathbb{R}^{d_h \times d_f}$, adding $d_h \times d_f$ per head
- Total per head: $d_f(d + d_h)$. With $d_f = d_h = d/H$: $\frac{d}{H}(d + \frac{d}{H}) = \frac{d^2}{H} + \frac{d^2}{H^2}$
- Total for H heads: $d^2 + d^2/H$
- Compared to QKV projection ($3d^2$) and output projection (d^2): the social field adds $\sim 25\%$ parameters to the attention layer (for typical H)

Gating adds bias terms (d_h per head = d total). Rotational adds a bilinear interaction, roughly $d_h^2 \times d_f$ per head, which can be expensive.

6.2 Computational Cost

The sequential bottleneck remains: the feedback loop requires position-by-position processing because position i 's deposit depends on the field state from positions $0, \dots, i - 1$.

Training. During training on sequences of length T , the forward pass requires a sequential scan of T steps. At each step: one matrix-vector multiply for the deposit ($O(d_f \times d)$), one for the modulation ($O(d_h \times d_f)$), and a vector add for the field update ($O(d_f)$). The attention computation itself ($O(T \times d_h)$ per position) dominates. Total cost is $O(T^2 d_h)$ same as standard attention, but the constant factor increases and parallelism across positions is lost.

Inference. In autoregressive inference, standard attention is already sequential. The social field adds only $O(d_f)$ work per position for the field update and $O(d_h \times d_f)$ for key modulation—negligible compared to the $O(T \times d_h)$ attention computation. **At inference time, the social field is essentially free.**

Potential acceleration. The sequential constraint during training could be relaxed via:

- *Chunked processing*: process chunks of C positions in parallel (using only the field state at the chunk boundary), then update the field. This trades accuracy for parallelism.
- *Truncated backpropagation*: detach the field from the computation graph every K positions, allowing parallel chunks of length K with approximate gradients.
- *Linear attention within chunks*: use linear attention (which supports parallel scan) for the field-dependent component, keeping softmax attention for the structured channel.

6.3 Length Generalization

The social field recurrence $\phi_i = \alpha \cdot \phi_{i-1} + \mathbf{d}_i$ is position-independent: it depends only on the previous field state and the current deposit, not on absolute position. This means the field dynamics are *identical* at position 50 during training and position 500 during inference—the recurrence does not know where it is in the sequence.

Combined with a position-independent attention mechanism (RoPE or no positional encoding), the entire dual-channel system is position-independent. The structured channel matches by content, the social channel routes by accumulated history, and neither requires knowing the absolute position. This is the architectural basis for length generalization.

Our prior work showed that even a scalar field (stigmergic attention) provides measurable length generalization on parity, despite using learned positional embeddings that break at unseen positions. With vector key modulation and position-independent encoding, the advantage should be substantially larger: the field carries more state (vectors vs scalars), modulates geometry (not just magnitude), and the attention mechanism itself still functions at OOD lengths.

7 Experimental Design

We propose experiments targeting the specific capabilities that dual-channel attention provides beyond standard attention: sequential state tracking and length generalization.

7.1 Tasks

Parity (1-bit state). The simplest state-tracking task, previously shown to benefit from scalar feedback. Dual-channel attention should show a larger effect because the vector field can encode parity as a *direction* in key space, not just a scalar magnitude.

Cumulative sum mod V ($\log_2 V$ -bit state). Previously failed with scalar field because the state exceeds scalar capacity. With $d_f = d_h = 12$ and $H = 4$ heads, the field has 48 continuous values of state—far more than the 4 bits needed for mod-16 arithmetic.

Bracket matching (counter + stack). Given sequences of open/close brackets with possible nesting, predict whether the sequence is balanced at each position. This requires a counter (nesting depth) and implicit stack (which bracket type to match). The field could encode depth as a vector magnitude and bracket type as a vector direction.

Evaluation protocol. For all tasks: train on sequences of length T , evaluate on $T, 2T, 3T, 4T$. Use RoPE or no positional encoding to isolate the field’s contribution to length generalization. Compare: standard attention, scalar stigmergic attention, and dual-channel attention with additive and gating modulation.

8 Scaling Considerations

The sequential bottleneck is the primary concern. At training time, the feedback loop forces sequential processing. For long sequences ($T > 1000$), this makes training prohibitively slow compared to standard parallel attention. However:

- At inference time (autoregressive generation), processing is already sequential—the social field adds negligible cost.
- For encoder models processing fixed-length inputs, the sequential constraint applies, but can be mitigated by chunked processing.

- The field could be applied to only a subset of layers (e.g., every 4th layer), reducing the sequential burden while maintaining most of the benefit.

Field dimension scaling. Setting $d_f = d_h$ means the field scales with model size automatically. A model with $d = 768, H = 12$ gets $d_f = 64$ per head, providing $12 \times 64 = 768$ values of persistent state. This is comparable to Mamba’s state dimension and should be sufficient for complex state tracking.

Gradient flow through the recurrence. The field recurrence $\phi_i = \alpha\phi_{i-1} + \mathbf{d}_i$ has gradient $\partial\phi_i/\partial\phi_j = \alpha^{i-j}$. For $\alpha = 0.95$ and $i - j = 100$: $0.95^{100} \approx 0.006$. Gradients through the field decay exponentially, potentially causing vanishing gradient problems for long-range dependencies. Possible mitigations:

- *Learnable retention*: let α be a learned parameter per head, initialized near 1.0
- *Multi-scale fields*: different heads at different retention rates, spanning timescales
- *Gated update*: $\phi_i = g_i \odot \phi_{i-1} + (1 - g_i) \odot \mathbf{d}_i$ with learned gate g_i , allowing selective persistence (cf. GRU)

Interaction with existing optimizations. Flash Attention exploits the fact that softmax attention is block-decomposable. Key modulation preserves this property—the modulated keys $\mathbf{k}'$ can be precomputed (for non-feedback variants) or computed per-chunk (for chunked feedback). KV caching at inference time extends naturally: cache the *modulated* keys $\mathbf{k}'_j$ along with values.

9 Discussion

9.1 The Individual/Social Decomposition

The core claim is that attention performs two conceptually distinct operations that benefit from separate mechanisms:

1. **Individual lookup**: “Given what I’m looking for (query) and what’s available (key), how relevant is each position?” This is *content-addressed* and *memoryless*—it depends only on the current position’s private state.
2. **Social routing**: “Given the collective processing history of the sequence, which directions in key space are worth attending to?” This is *history-dependent* and *collectively derived*—it comes from the social field, not from any single position.

Standard attention collapses these into a single Q/K dot product. Learned positional encodings partially separate them (position provides structure-dependent routing) but are static and length-limited. The social field provides *dynamic, experience-derived, length-independent* routing that adapts to each sequence individually—the collective intelligence of the sequence’s own processing history.

9.2 What the Social Field Learns

In biological stigmergic systems, environmental traces encode three types of information:

1. **Recency**: recent deposits are stronger (exponential decay). In the social field: α^{i-j} weighting.

2. **Frequency:** frequently productive paths accumulate more signal. In the social field: positions that consistently produce useful outputs accumulate deposits.
3. **Direction:** field gradients indicate which way to go. In the vector social field: the field encodes a *direction* in key space, not just a magnitude.

The scalar stigmergic field of our companion paper captured recency and frequency but not direction. The vector social field captures all three, which is why we expect qualitatively different behavior: the social channel can steer attention through a learned geometry, not merely amplify or suppress it.

9.3 Limitations and Open Questions

- **Training speed:** The sequential bottleneck is real and significant. Chunked approximations trade accuracy for speed, and the optimal tradeoff is unknown.
- **Task generality:** State-tracking tasks are constructed to favor recurrent mechanisms. Whether the social field helps on natural language tasks—where the “state” is diffuse and high-dimensional—is unknown.
- **Optimization difficulty:** The feedback loop creates long gradient paths through the recurrence and the attention mechanism. This may make training unstable or slow to converge. The seed-dependent bimodality observed in scalar experiments suggests the optimization landscape has multiple basins.
- **Redundancy with existing mechanisms:** Multi-layer transformers already build up sequential context through layer composition. The social field may provide redundant state-tracking that deeper networks achieve anyway.

9.4 Isn’t the Residual Stream Already a Social Field?

A natural objection: deep transformers already exhibit the three critical ingredients, so why add another mechanism? The residual stream *is* a shared medium. Each layer deposits into it (injection). Softmax creates winner-take-all dynamics that compound across layers (amplification). LayerNorm rescales activations, preventing runaway accumulation (erosion of a sort). The critic concludes: the transformer is already a stigmergic system, and the social field reinvents what depth provides for free.

This objection is half right, and engaging it precisely clarifies what the social field actually contributes.

The residual stream is social across layers, not across positions. At any given layer, each position’s attention computation is memoryless with respect to other positions’ *processing outcomes*. Position i sees the raw keys and values of positions $j < i$, but it does not see what positions $j-1, j-2, \dots$ found useful when *they* attended. The residual stream carries information forward through depth (layers) but not through sequence (positions within a layer). The social field carries information forward through the sequence dimension—precisely the dimension where standard attention has no memory. This is why state-tracking tasks (parity, counting, register maintenance) are hard for standard transformers: they must emulate sequential accumulation using combinatorial depth $\times$ width, rather than having an architectural primitive for it.

Depth is fixed; the social field scales with sequence length. A 12-layer transformer has exactly 12 opportunities for inter-layer coordination, whether the input is 10 tokens or 10,000. The social field accumulates deposits proportional to sequence length—it has *more* collective memory for longer sequences. This is the architectural basis for length generalisation: the field’s recurrence is position-independent and naturally extends to unseen lengths, while depth-based state emulation is capacity-bounded.

The residual stream does not modulate the geometry of attention. Information in the residual stream enters the next layer’s attention through generic QKV projections. The stream changes what gets projected, but through a learned linear transform that treats the residual vector as undifferentiated input. The social field specifically modulates *keys*—it operates in the same representational space as the Q/K dot product, reshaping the matching geometry rather than merely providing different input features. This is the difference between “information flows forward” (trivially true of any deep network) and “a decaying field reshapes which directions in key space are matchable.”

The uncoupling constraint: where the ant model builds intuition. The most non-obvious distinction concerns erosion. LayerNorm’s rescaling is *field-correlated*: it responds to the magnitude of the activations it normalises, shrinking large values and expanding small ones. Our lattice ant study provides direct intuition for why this matters. Of nine model variants we tested, every variant whose decay was correlated with the pheromone field—follow-probability modulation, habituation, sensory saturation—destroyed the critical ridge entirely. Only variants with uniform, field-independent decay preserved edge-of-chaos dynamics. The mechanism is intuitive once seen in the ant system: field-correlated perturbations feed back into the amplification loop, creating a self-cancelling cycle that prevents the delicate balance between growth and decay that defines criticality. LayerNorm does exactly this—it is a field-correlated erosion that actively counteracts the very accumulation patterns that would otherwise self-organise. The social field’s uniform α decay, by contrast, satisfies the uncoupling constraint. This distinction would be purely theoretical without the ant model; the lattice simulations make it concrete and testable.

The honest assessment. Multi-layer transformers *do* track state through depth—modern large models handle parity, counting, and bracket matching. But they do so by learning to emulate sequential state maintenance, not by having a native mechanism for it. The social field makes sequential state explicit. Whether this leads to faster learning, better generalisation, or mere redundancy is exactly what the experimental programme (section 7) is designed to test. The architectural argument is that three specific properties—positional memory, uncoupled erosion, and geometric modulation—are absent from the residual stream and present in the social field. Whether those properties *matter* computationally is an empirical question.

9.5 Lessons from the Ant Model: Uncoupled Fields and Criticality

A striking result from our lattice ant study is that *edge-of-chaos dynamics require the decay (erosion) channel to be uncoupled from the pheromone field itself*. Of nine model variants tested, only those with uniform, field-independent decay produced a critical ridge; every variant that introduced disorder correlated with the pheromone concentration—follow-probability modulation, habituation, sensory saturation, coupled deposition—destroyed criticality by feeding field-dependent noise back into the amplification loop. In short: the hard problem is not *producing* edge-of-chaos dynamics but *not destroying* them with field-correlated perturbations.

This has a direct architectural implication for dual-channel attention. The social field’s decay parameter α should act as a *uniform* exponential forgetting factor, independent of the

field’s current magnitude. If decay were made content-dependent (e.g. stronger erasure where the field is large, or gated by the field’s own values), the ant model predicts that the system would lose its ability to self-organise near criticality. The separation of concerns—individual attention handles content-dependent computation, while the social field evolves under content-*independent* erosion—may therefore be essential, not merely convenient. Uniform decay preserves the directed-percolation structure that enables critical dynamics; field-coupled decay collapses it.

9.6 Reflections: Sociality as Structure Discovery

A useful way to distinguish the individual and social channels is by what kind of knowledge they encode. Individual lookup is literal: given this query, find the matching key—a direct content-addressed retrieval. The social field does something qualitatively different. By accumulating deposits across positions, it builds a *compressed record of which directions in key space were collectively productive*. This is not a lookup table but a structural prior: things shaped like *this* tend to benefit from rules shaped like *that*. Where individual attention asks “what should I do here?”, the social field asks “what kind of situation is this, and what has worked for situations like it?”

This is, in a sense, the hard version of the problem. Finding literal associations is cheap; discovering that two superficially different contexts share hidden structure—that they are instances of the same *metaphor*—requires the system to compress over its own processing history. But the hypothesis motivating this architecture is that the world is full of such reusable structure, and this is precisely what makes it learnable. Biological systems exploit this ruthlessly: evolution builds on older solutions without relearning from scratch, reusing regulatory motifs, signalling cascades, and body plans across wildly different contexts. The social field is a minimal mechanism for the same principle—finding and propagating structural regularities rather than storing each situation individually.

10 Conclusion

We have argued that the three ingredients governing critical phase transitions in the directed percolation universality class—amplification, injection, erosion—appear in both ant trail formation and biological neural plasticity mechanisms (synaptic tagging, volume transmission, activity-dependent myelination). The ant trail model is the simplest system that makes these ingredients and their phase transition visible; we use it as an illustration, not a metaphor.

Our companion paper [13] tested the simplest possible import of these ingredients into attention: a scalar pheromone field biasing logits. The results confirmed that the feedback loop matters (deposits must come from output, closing the amplification loop) but showed that a scalar field decoupled from key geometry is too weak—softmax normalization absorbs most of the signal. This motivated the present architecture.

Dual-channel attention decomposes the attention mechanism into individual lookup (content-based Q/K matching) and social routing (vector field key modulation). The social field provides intrinsic sequential memory— $H \times d_f$ continuous values of persistent, decaying, content-dependent state per layer—without introducing a separate recurrent pathway. It operates in the same representational space as Q/K matching, modulating the *geometry* of attention rather than its magnitude.

The deeper question is whether criticality plays a functional role in learned sequence processing—whether there exists a regime where the social field’s feedback dynamics self-organize to a critical point, and whether this criticality confers the same computational advantages (maximal dynamic range, information transmission, sensitivity) observed in biological neural systems at criticality. The architecture is designed to make this question empirically testable.

A Research Notes: Unifying Learning and Inference

A.1 The Separation in Standard Attention

In a vanilla transformer, there is a clean division of labour: *training* learns the projections W_Q , W_K , W_V —essentially a filing system for where to put information and how to retrieve it. At *inference* time, those projections are frozen, and the model executes retrieval. “In-context learning” in standard transformers is somewhat misleadingly named: the model is not learning anything during the forward pass. It is recognising patterns it already learned to recognise during training. The weights do not change. It is retrieval all the way down.

A.2 What the Social Field Changes

The social field actually *does* modify itself during the forward pass. As it processes positions 1, 2, 3, ..., it accumulates deposits that reshape the attention geometry for all subsequent positions. Position 50 attends in a key space that has been sculpted by the collective processing outcomes of positions 1–49. This is genuinely different: the field is building temporary structure *during inference* that did not exist before and was not hardcoded in the weights.

Standard attention learns *where to file*, then retrieves. The social field learns *what structure exists in this particular sequence* as it processes it, and that structure modifies how it processes what comes next. The deposit–accumulate–modulate loop is functionally a learning algorithm operating at inference time.

A.3 Three Timescales

The deeper point concerns what “learning” means at different timescales:

1. **Training** (slow): learn W_Q , W_K , W_V , W_{deposit} —the rules of the game.
2. **Social field accumulation** (fast): discover the structure of *this particular sequence*—the shape of this game.
3. **Individual attention** (instantaneous): make a single retrieval given the current state.

Standard transformers have only the slow and instantaneous timescales. The social field introduces the *middle* timescale—slower than a single attention operation, faster than weight updates. This is exactly the timescale that is missing, and it is the timescale at which biological plasticity operates: synaptic tagging, short-term potentiation, neuromodulatory volume transmission—all happening during “inference” in the brain.

A.4 The Biological Parallel

Biological brains do not have a train/infer separation at all—plasticity is continuous. What they *do* have is multiple timescales of plasticity: fast (ion channel states, milliseconds), medium (synaptic tagging, minutes to hours), slow (structural plasticity, days to weeks). The social field is a minimal version of this: it adds one intermediate plasticity timescale to an architecture that otherwise only has “frozen weights” and “instantaneous computation.”

The presupposition that the world has countable order is important here. The social field can only help if the structure it discovers early in a sequence is *predictive* of structure later in the sequence—which is exactly the assumption that the world has reusable regularities. If every position were independently random, the social field would accumulate noise. That it accumulates *signal* is a bet on the world being structured, and this is the same bet that makes evolution possible: old solutions remain useful because the world’s structure is stable enough to reuse.

A.5 A Testable Provocation

The social field brings learning and inference closer, but they are not fully unified—the field resets between sequences. True unification would mean the field persists and the system improves across sequences, not just within one. The social field is more like “inference that remembers its own recent history” than “learning.” But this may be exactly the right intermediate step, and it is notable that biological short-term plasticity also resets: synaptic facilitation decays, working memory clears—and yet it is clearly doing something different from pure retrieval.

The really provocative version of this claim: *the reason transformers need so many parameters and so much training data is that they are forced to compile all sequential structure into static weights, because they lack a mechanism for discovering structure at inference time.* The social field could, in principle, let a smaller model with less training discover structure on-the-fly that a standard transformer would need to have memorised. This is testable, and is the direction we are chipping away at.

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